

deviation. DD receives current together with voltage, temperature, cumulative charge, derivatives, and temporal context. Its smaller observed penalty is therefore consistent with a lower dependence on one systematically scaled channel.

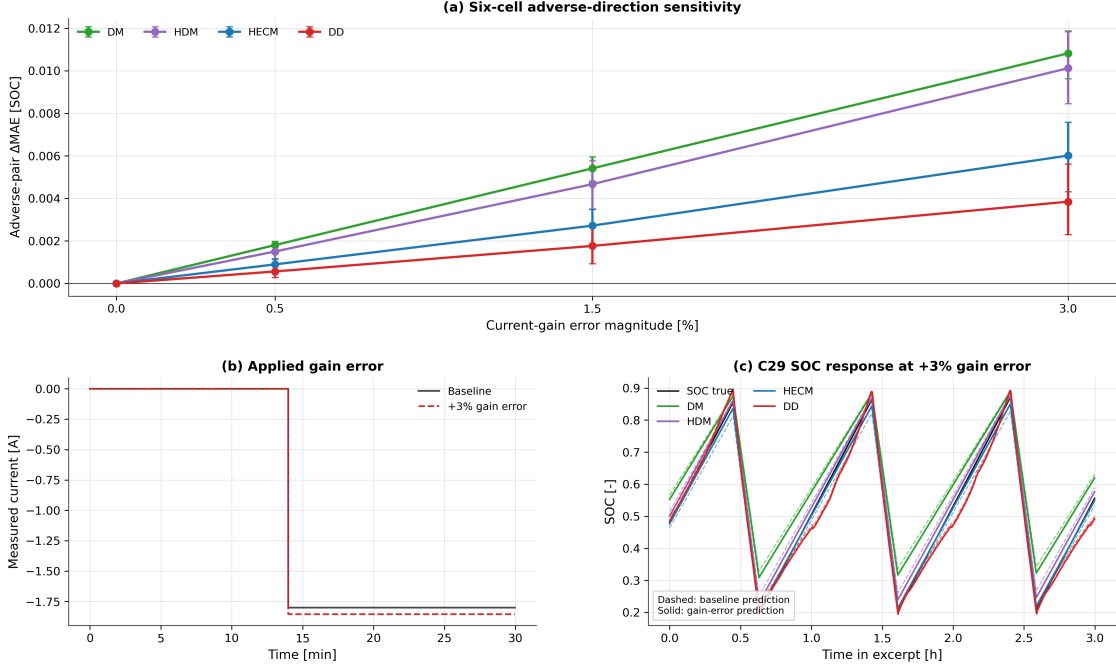


Figure 6.5: Current-gain sensitivity. Panel (a) reports six-cell adverse-direction Δ MAE for every matched signed gain pair with hierarchical 95% confidence intervals. Panel (b) compares representative current before and after the +3% gain error. Panel (c) shows SOC trajectories from the representative high-load holdout trajectory over three charge and discharge intervals.

The continuous high-load lifecycle analysis in Figure 6.6 complements the independent 24-h windows. The gain-error contribution changes over life because signed Coulomb-counting error interacts with current direction and operating state. The penalty often grows between full charges, while a full-charge anchor can reduce or restructure it. The process is therefore neither unrestricted monotonic drift nor a complete reset. Across this single full-life replay, the adverse 3% gain error raises MAE by 0.0102 for DM, 0.0072 for HDM, 0.0037 for HECM, and 0.0048 for DD. HECM has the smallest contribution in this trajectory. This mechanism-focused diagnostic does not replace the six-cell window comparison.